Duckophila: a whole-brain *Drosophila* connectome model in closed loop with a bipedal robot

Technical report and preprint draft

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Abstract

We connect the complete connectome-derived leaky integrate-and-fire model of the adult *Drosophila melanogaster* brain (FlyWire, 138,639 neurons, 54.5 million synapses; Shiu et al. 2024) to a small bipedal robot, Pollen Robotics' microduck, running its production control software against a MuJoCo simulation of its body. The robot's time-of-flight depth sensor is translated into input to the fly's looming-sensitive visual projection neurons (LPLC2, LC4); its camera, through dense optic flow, into input to the fly's elementary motion detectors (T4/T5), each placed at its own visual column using FlyWire's column map. Motor commands are read out from identified descending neurons. Without any tuning of the connectome, the model (i) fires its giant fibre (DNp01) to objects approaching the robot's face, on the side the object approaches from, and (ii) produces a syndirectional optomotor response in DNp15/DNHS1 that turns the robot's head—and, through a coupling rule of ours, its walking body—with a rotating striped drum, at a closed-loop gain of 0.3–0.4. To run the model in closed loop we reimplemented it as a stepped simulator that reproduces the reference Brian2 implementation spike for spike and runs faster than real time on one CPU core, using an exact closed-form bound that lets neurons unable to fire before their next input remain unsimulated. We report what the connectome contributes, what our hand-written sensory and motor interfaces contribute, and several failure modes found on the way: a dropped-input rule of the reference model, scrambled retinotopy destroying LPLC2 selectivity, the robot's own gait dominating its optic flow, and a field of view too narrow for the canonical steering neuron DNa02. Code, experiments and a recorded demonstration are open source.

1 Introduction

Complete synaptic wiring diagrams of an adult insect brain now exist. FlyWire reconstructed the whole brain of a female *Drosophila* [1], annotated every neuron with a cell type [2], and mapped the visual system to its retinotopic columns [3]. Shiu et al. [4] showed that a leaky integrate-and-fire (LIF) network built directly from this connectome—every synapse at one shared weight, its sign taken from the predicted neurotransmitter—predicts sensorimotor pathways that were then confirmed experimentally, for example from sugar-sensing gustatory neurons to the proboscis motor neuron MN9.

That model has so far been driven by prescribed stimulation and read out offline. Here we ask a different question: what does it do when its sensory neurons are driven by a robot's sensors, its descending neurons move that robot, and the consequences of those movements come back through the sensors—in closed loop, in real time? The robot is the microduck, a 15-servo bipedal robot from Pollen Robotics with an 8×8 time-of-flight (ToF) depth sensor and a head camera, controlled by reinforcement-learned locomotion policies. We do not replace the robot's locomotion: the fly brain sits above it, as a behaviour source, the way descending neurons sit above the fly's ventral nerve cord.

Our contribution is an engineering study with three parts: (1) a stepped, exact and faster-than-real-time reimplementation of the Shiu et al. model; (2) sensory interfaces that respect the fly's own anatomy—in particular a retinotopic placement of every T4/T5 neuron whose orientation is derived from the connectome itself; and (3) closed-loop experiments in a simulated twin of the robot that runs the robot's production daemons. Throughout, we separate what the connectome does from what our interfaces do, because that boundary is what any claim about "a fly brain controlling a robot" depends on (Table 1).

Table 1: What is the connectome's and what is ours. The brain model is not trained, tuned or edited.

From the connectome (unchanged)	Hand-written interfaces (ours)
Every neuron, synapse and sign; the LIF dynamics of [4]	ToF → LPLC2/LC4 Poisson rates (inverse time-to-contact)
Which neurons respond to looming, which to rotation	Camera → optic flow → T4/T5 rates by preferred direction
Whether, and on which side, the giant fibre fires	The shape of the startle (sound, gaze, crouch, step back)
The sign and size of the DNp15/DNa02 left–right difference	Gains, smoothing, clamps; the body following the head
Where each T4/T5 looks (FlyWire columns); front and up, from the wiring	Approach-speed gate; efference copies of the robot's own motion

2 Methods

2.1 Brain model and a stepped reimplementation

We use the FlyWire 783 connectome as distributed with [4] (138,639 neurons, 15,091,983 connected pairs, 54,492,922 synapses) and the model's equations and constants unchanged: membrane time constant 20 ms, synaptic time constant 5 ms, rest and reset −52 mV, threshold −45 mV, refractory period 2.2 ms, synaptic delay 1.8 ms, 0.275 mV per synapse, time step 0.1 ms, and Poisson stimulation as a voltage kick of 250 × 0.275 mV.

The reference implementation in Brian2 [5] rebuilds the network for every run, which is appropriate for one-second optogenetic trials but not for a sensorimotor loop that needs tens of milliseconds of brain time per sensor frame (a 20 ms step cost 0.2 s to 0.4 s of wall time; a long run, 2.6× real time). Our stepped simulator (`flysim.py`, numba) holds the state between calls and follows Brian2's schedule within each step: state update, threshold, synaptic propagation and Poisson input, reset.

A semantic detail of the reference. In the reference model v and g are declared `(unless refractory)`. Brian2 implements this as a *conditional write*: any write to these variables, including synaptic input, is discarded while the neuron is refractory. Input arriving during refractoriness is therefore lost, not accumulated. An implementation that banks it is plausible-looking but wrong: every downstream neuron fired ∼20% too fast in ours until this was matched. We found it by replaying the reference's own input spike times into our simulator and comparing every downstream spike.

Exact sparsity. Under strong visual drive ∼50,000 neurons sit just off rest, and stepping them made the brain four times slower than real time; parallelising the per-step pass across cores made it slower still (thread-region launch every 0.1 ms). Between inputs the model is linear, so the highest a neuron's voltage can still reach without further input has a closed form:

$$v_{\max} - v_0 \le \max(u, 0) + P_g \max(g, 0), \qquad P_g = \max_t \frac{\tau}{\tau - T}\left(e^{-t/\tau} - e^{-t/T}\right) \approx 0.157, \tag{1}$$

with $u = v - v_0$, T the membrane and τ the synaptic time constant. A non-refractory neuron whose bound is below threshold cannot spike before its next input; it becomes *dormant* and is not stepped. Its next synaptic arrival or Poisson event wakes it, and its state is advanced analytically over the gap (the per-step propagator raised to the gap, from a lookup table). Poisson input is drawn as geometric inter-event gaps per frame rather than a Bernoulli trial per neuron per step, which is the same process. The result remains spike-for-spike identical to Brian2 under replay (13,607 of 13,607 spikes) while cost follows synaptic events instead of neurons × steps.

2.2 Robot, simulator and interface

The microduck's production daemons (`robotd`, `tofd`, `mediad`) run unmodified against a MuJoCo [6] model of its body, with only the servo bus, the ToF sensor and the camera replaced by the simulator (`scripts/duck-sim`). Our bridge reads the robot's own interfaces: ToF frames at 15 Hz (`tof.stream`), camera frames at 30 Hz (`media.frame`), and the robot's state at 50 Hz (`robot.state`: head forward kinematics, IMU orientation, odometry, executed twist). It writes only intents the robot already accepts—`robot.move`, `robot.head`, `robot.look`, `robot.pose`, `robot.sound`. The brain advances by exactly the time between ToF frames, with the latest camera-derived rates.

A stimulus prop (a sphere that can be thrown at the robot) and a rotating drum of 48 alternating black and white panels (0.5 m radius, a 15° spatial period) are added to the simulator without modifying it: a generated copy of the scene, and a Python start-up hook that moves the two mocap bodies.

The twin had to walk first. The twin as shipped loaded an obsolete walking policy on a robot model without joint backlash, and did not walk: 4 cm in 8 s at 0.15 m/s commanded. The current policy (velstand) was trained with backlash and stands still without it; with both corrected the twin walks (0.06 m/s for 0.15 m/s asked) and turns (50 ř/s for 46 ř/s asked on the spot). This fix has been submitted upstream. The residual shortfall in forward speed, a dead band below ∼0.55 rad/s of yaw on the spot, and an occasional failure to take a first slow step are properties of the twin that our controllers have to live with.

2.3 Startle: depth sensor to looming detectors

The giant fibre (DNp01, one per side) receives its two largest excitatory inputs from LPLC2 and LC4 (10.6% and 8.6% of the left GF's annotated input synapses; 12.1% and 8.4% of the right's), consistent with their established roles in looming detection [9, 10, 8]. For each ToF zone we estimate inverse time-to-contact $\tau^{-1} = -\dot{d}/d$ from a line fit over four frames (0.27 s); a zone drives the LPLC2 and LC4 neurons assigned to it (left four columns to the left optic lobe, round-robin within a side, ∼3 LPLC2 and ∼1.5 LC4 per zone) at a saturating rate up to 150 Hz. Two filters are ours and labelled as such: zones whose return is the floor are excluded, using the robot's own floor test with a 3° pose margin; and drive is gated on approach speed above 0.4 m/s, a duck choice that loosely parallels the giant fibre's preference for fast looming [9]. A giant-fibre spike triggers a startle assembled from existing intents, with a 2 s refractory period.

2.4 Optomotor reflex: camera to motion detectors

Retinotopy from the connectome. Each eye has ∼750 T4 and ∼750 T5 neurons of each of four preferred directions (a front-to-back, b back-to-front, c up, d down [7]). FlyWire's column map [3] assigns 11,822 of them to 796 hexagonal columns (395 left, 401 right) with axial coordinates (p, q). Nearest neighbours differ by $(p{+}1, q{+}1)$ or by one coordinate, which gives physical coordinates $X = \sqrt{3}\,(q - p)/2$, $Y = (p + q)/2$ and a roughly round eye.

The map does not say which direction is anterior or dorsal. We derive both from the wiring (Fig. 3). LPLC2's four dendritic arms each take the T4/T5 subtype whose preferred direction points away from the cell's centre [8]; averaging, over all LPLC2 of an eye, the position of each subtype's inputs relative to the cell's input centroid places T4a/T5a at $\Delta X = -2.7$ and -2.9 columns (left and right eye), T4b/T5b at $+2.1$ and $+1.9$, T4c/T5c at $\Delta Y = +1.7$ and $+1.9$, and T4d/T5d at -2.1 and -2.2. Hence $+X$ is anterior and $+Y$ dorsal in both eyes. Independently, the dorsal and ventral HS cells (HSN, HSS) take their T4/T5 input at higher and lower Y respectively, confirming the vertical axis, and the equatorial HSE sets elevation zero. With an interommatidial angle of 4.8° and the frontmost column placed 10° across the midline, the eyes span $-10°$ to 131° in azimuth and $-84°$ to 66° in elevation.

Flow to rates. The camera sees 45°×73° straight ahead. Only T4/T5 neurons whose direction falls in that view—1,424 of 11,822—receive input: dense Farnebäck optic flow [15] at quarter resolution, averaged over one column's extent at the neuron's position, projected on its preferred direction, rectified and mapped to a saturating Poisson rate (≤60 Hz). Image left is the robot's left (verified with a placed object).

Efference copies. Image motion the robot causes deliberately is predicted and subtracted before it reaches the fly: the head's idle look-around, and walking. Walking turned out to be dominated not by translation (floor flow ∼10 ř/s at 0.08 m/s) but by the gait's pitch and roll (30 ř/s to 80 ř/s). We predict both per pixel from the robot's state—floor translation from odometry on pixels whose ray meets the floor, rotation from the camera's world orientation between states with the vertical component removed (yaw slip is the reflex's own signal)—and subtract the prediction weighted by local image texture, since a flow estimator, like a T4 neuron, reports no motion on an edgeless surface. The prediction matches a controlled head nod with $r = 0.98$; during walking it correlates with the measured flow ($r = 0.90$) only when each frame is paired with the state at its own capture time.

Motor readout. DNp15 (DNHS1 [11]) left–right difference, smoothed (0.15 s), integrates into head yaw. For the body we tested DNa02 [12] directly (`-steer dna02`) and a head-led coupling of ours (`-steer body`): the walking body (0.1 m/s) turns at $k\cdot$ head yaw while the head counter-rotates by the trunk's measured yaw rate, so gaze is held.

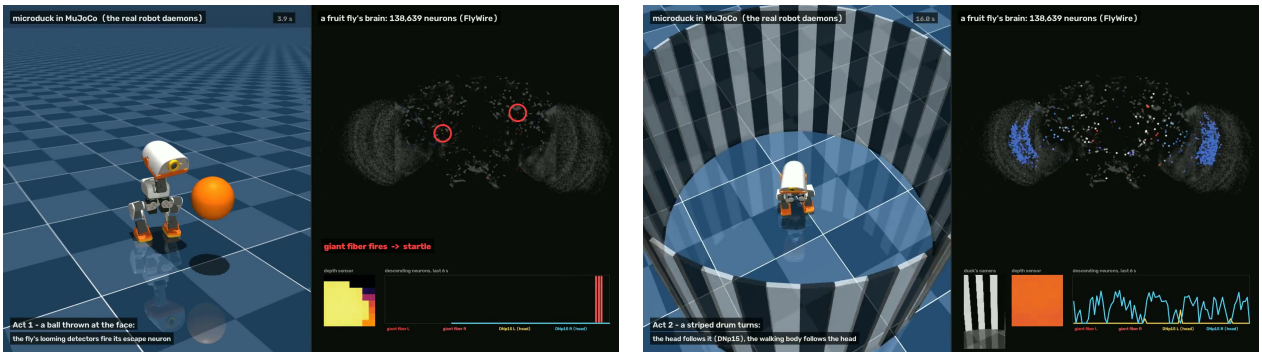


Figure 1: Frames from the recorded demonstration (`media/duckophila.mp4`). Left halves: the robot in MuJoCo, re-rendered from its recorded pose. Right halves: the brain, every neuron at its FlyWire position seen from the front, lit in its class's colour when it spikes. **Left:** a ball reaches the robot's face; both giant fibres are ringed in red as they fire. **Right:** in the turning drum, T4/T5 neurons in both optic lobes (blue) and DNp15 respond.

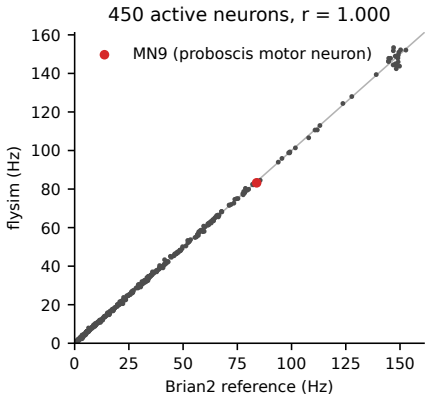


Figure 2: The stepped simulator against the Brian2 reference: mean firing rates over 30 trials with the right sugar-sensing neurons at 150 Hz (FlyWire 630, as in [4]).

3 Results

3.1 The stepped model reproduces the reference

With the sugar-sensing gustatory neurons driven at 150 Hz for 30 one-second trials, firing rates of the 450 active neurons correlate with the Brian2 reference at $r = 0.9998$; MN9 fires at 83.2 Hz against 83.9 Hz (Fig. 2). Under replay of the reference's own input spikes, all 13,607 downstream spikes are identical in neuron and time step. On one core of an Apple M5 Max, one second of brain takes 0.04 s under that stimulus, 0.38 s under drum-like drive of 3,000 T4/T5 at 30 Hz, and 0.78 s with all 12,246 T4/T5 at 30 Hz; in the closed loop, a 67 ms frame costs 4 ms to 7 ms.

3.2 Startle

Driving LPLC2 and LC4 of one side directly, the giant fibres are silent at rest and fire in a graded way with drive rate and with the fraction of the population driven, more strongly ipsilaterally; at higher drive the contralateral DNa01/DNa02 and, weakly, the moonwalker descending neurons (MDN [13]) are recruited. On synthetic ToF scenes ray-cast with the sensor's beam geometry and noise, all 21 trials (7 scenes × 3 seeds) were classified correctly: a hand at 1 m/s or a ball at 1.5 m/s fired the giant fibre 0.13 s to 0.27 s after onset (the hand still ∼0.6 m away), while sensor noise, walking towards a wall at 0.15 and 0.3 m/s, a ball passing at 40 cm and a hand at 0.2 m/s produced no giant-fibre spike; 0.4 m/s is the edge. In the simulated robot, in the recorded demonstration, each of three balls at 1.5 m/s from centre, left and right produced a startle within a second, the left ball through the left giant fibre and the right ball through the right; a hand at 0.2 m/s produced none.

Two failure modes were found and fixed in the interface, not the brain. A standing robot's idle head motion makes ∼45 of 64 depth zones see an approaching and receding floor; without a floor filter 30% of quiet frames drove the looming detectors. Near the horizon, a 2° to 3° discrepancy between the kinematic

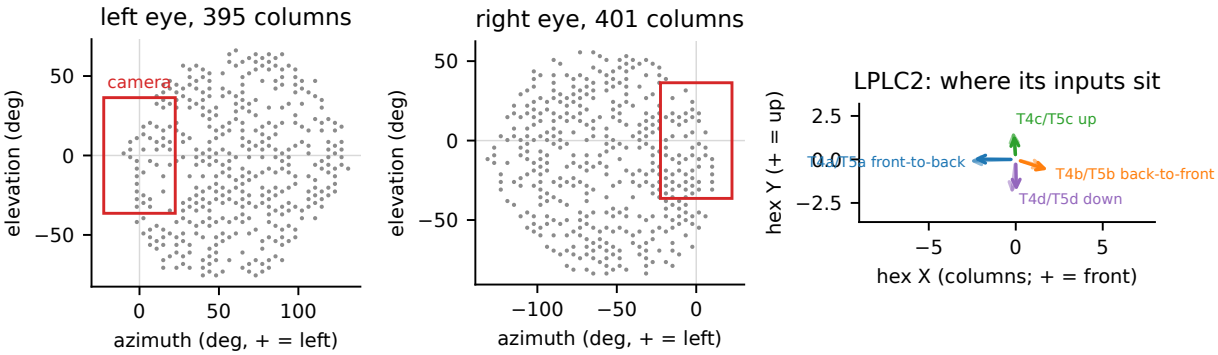


Figure 3: Retinotopy from the connectome. **Left, centre:** viewing directions of the 395 and 401 columns of each eye, from FlyWire's column map with orientation derived from the wiring; red, the camera's field of view. **Right:** mean offset of each T4/T5 subtype's input to LPLC2 from the LPLC2 neuron's input centroid (solid: left eye; faint: right). Front-to-back inputs sit posterior, up inputs dorsal—the outward-motion arrangement of a looming detector—which fixes the eye's axes.

model and the rendered sensor pose decides whether a grazing return is floor; without a pose margin, the robot startled 5 times in 25 s in an empty room, with it none.

3.3 Optomotor reflex

In the connectome. Driving T4a/T5a of one eye and T4b/T5b of the other—the pattern a yaw rotation of the world produces—activates the ipsilateral HS cells and contralateral H2 as in the fly, and flips the left–right activity of DNa02 with the direction of rotation, in the sense of a syndirectional (following) turn under the interpretation of [12]: world turning right, DNa02 right 28 Hz versus left 5 Hz; world turning left, left 69 Hz versus right 5 Hz. DNa01 does not flip cleanly. The strongest-flipping descending neuron is DNp15 (DNHS1).

Retinotopy is necessary. With T4/T5 neurons assigned to image regions at random, wide-field drum motion reached LPLC2 scrambled and fired the giant fibre 7–11 times per run. With column-based placement, rotation produced no giant-fibre activity in synthetic tests or in the closed loop (one startle in 29 s of drum, from the ToF at start-up).

The camera's field of view limits the readout. With the column map, a whole-eye yaw rotation flips both DNa02 and DNp15 (Fig. 4). Restricted to the neurons whose direction falls in the camera's view, DNp15 still flips strongly (51 Hz left vs. 0 Hz right; 0 vs. 86 Hz), but DNa02 does not change sign: right exceeds left in both directions, by 2.7 Hz and 11 Hz. Rotation over the frontal 30° of the eye at all elevations is sufficient for a DNa02 flip; the camera's limited elevation range is not. Likewise, no expansion stimulus within the camera view, up to 400 ř/s, drove LPLC2 appreciably; looming therefore remains the ToF's task.

Closed loop. With DNp15 driving head yaw, the robot's head follows the drum in both directions as a triangle wave at 8 ř/s to 12 ř/s for a 30 ř/s drum (closed-loop gain 0.3–0.4: partial compensation, as optomotor responses typically show) and holds still when the drum is still; higher gains oscillate, given ∼0.3 s of loop latency. With the head-led body coupling, the walking robot turns with the drum at a mean of +5 and −9 ř/s (peaks ±15 ř/s to 20 ř/s) against an intrinsic gait drift of −4.6 ř/s. Without the walking efference copy, three startles occurred in 200 s of walking in the turning drum; with it, none, with unchanged steering.

4 Discussion

The connectome, unmodified and untrained, produced two recognisable fly behaviours when its sensory neurons were driven by a robot's sensors: a lateralised escape decision to fast approaching objects, and a

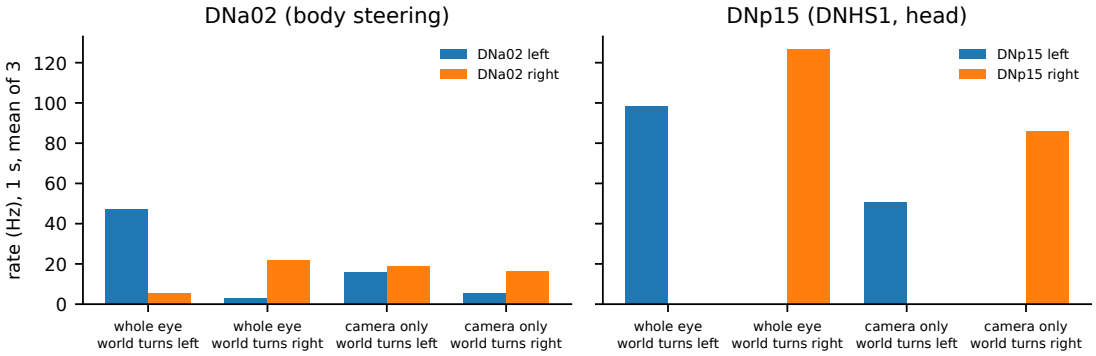


Figure 4: Steering descending neurons under yaw rotation (40 ř/s), with every T4/T5 driven (whole eye) or only those looking into the camera's view. DNp15 flips in both cases; DNa02 only with the whole eye.

following optomotor response. Neither was wired in. What was wired in is listed in Table 1, and most of the effort went there: the biological plausibility of the input decided whether the connectome's computation could be expressed at all. Scrambled retinotopy did not just add noise, it produced false escape decisions, because LPLC2's selectivity is a property of where its inputs sit. The robot's own movement dominated its visual input, as it does for animals, and needed an efference copy before the fly's motion pathway could be used while walking [14].

The clearest limitation is sensory. A single forward camera covers a small part of a fly's visual field, and the canonical steering neuron DNa02 [12] did not respond to rotation within it. The head-coupled neuron DNp15 did, so we steer the head with the fly's output and the body with a coupling rule of ours. Whether a wider camera, or several, recover DNa02-based steering is a direct experimental question the setup can answer.

Other limitations are those of the model and the platform. The LIF model of [4] uses one synaptic weight for all connections, has no gap junctions (the giant fibre's electrical synapses are absent), no plasticity and no neuromodulation, and treats neurotransmitter predictions as signs. T4 and T5 are driven alike, without an ON/OFF split. The robot is simulated; its gait reaches 40% of commanded speed and the camera/state timing measured in simulation will differ on hardware. Our gains and gates are engineering choices, stated as such. We make no claim that the robot behaves like a fly beyond the two reflexes measured.

Natural next steps are pursuit through the courtship circuit (LC10 → P1) as a "follow me" behaviour, valence learning in the mushroom body with plasticity at Kenyon-cell output synapses, and the physical robot, with the brain on a laptop and the robot's camera streamed over WebRTC. Related efforts embody connectome-constrained or neuromechanical fly models in simulated bodies [16, 17]; here the body is a different animal's, and the interesting part is the translation between them.

Reproducibility

Everything above is a script in `https://github.com/first-it-consulting/duckophila`: `experiments/` (each figure and number), `paper/figures.py` (every figure, from data), `demo.sh -video` (the recorded demonstration). The FlyWire data are fetched under their own terms. Hardware: Apple M5 Max, macOS; brain on one core.

Acknowledgements

The connectome and annotations are the work of the FlyWire consortium; the brain model is that of Shiu et al.; the robot, its software and its simulator are Pollen Robotics'. The FlyWire annotations used are version ≥3.2, for which the annotation authors also ask to cite Berg et al. (2026) and Tastekin et al. (2026). The implementation, experiments and this report were developed with substantial assistance from an AI coding assistant (Anthropic's Claude), directed and reviewed by the author.

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